Founder Memo: Reducing COD RTO Without Killing Growth

Date: 30 October 2026 **Subject:** Pilot proposal for COD order decisions

Recommendation. Start with a cautious, measurable pilot: ship low-risk orders normally; ask for WhatsApp confirmation on moderate-risk or data-quality-flagged orders; offer a small prepaid incentive for elevated-risk orders; and send the highest-risk orders to manual review. Do not automatically cancel high-risk orders at launch. This protects Operations from avoidable RTO costs without turning a model score into an irreversible decision for a potentially genuine customer.

What the supplied data says

RTO baseline. The sample has 2,018 rows; 2,017 have a final DELIVERED/RTO outcome after normalizing status text. There are 573 RTOs, or 28.4% of labeled orders—close to the brief's ~28% COD RTO baseline.

First-time orders are not "safe". Among 1,517 rows with zero prior orders, 30.3% ended in RTO. This is a reason not to assume first-time customers are low-risk, but it is also not a reason to cancel them: confirmation is a lower-friction intervention.

Prior RTO history is a useful signal, but small. Orders with `customer_prior_rto >= 2` had 26 RTOs among 34 rows (76.5%). The segment is small, so it should trigger confirmation/manual review rather than a blanket cancellation rule.

Address quality deserves attention. 144 labeled rows have addresses shorter than eight characters, including 41 rows whose address is a placeholder such as ".", "-", "na" or "test". These checks are cheap and deterministic; they should trigger confirmation before a dispatch decision.

Data quality is a deployment risk. 60 rows are involved in duplicate order IDs (30 ID values appear more than once); there are 12 missing pincodes and 4 missing cities in the raw file. Resolve whether repeated IDs are genuine duplicate events before using this sample for financial claims or production training.

Recommended rules and financial trade-off

Initial, editable risk bands: below 18% → ship as-is; 18–35% → WhatsApp confirmation; 35–55% → prepaid nudge; 55%+ → confirmation plus manual review. Obvious data-quality flags (placeholder/very short address, suspicious name, invalid phone length, or zero-value order) trigger confirmation regardless of model score. Auto-cancel remains off by default. These are pilot thresholds, not optimized cutoffs; adjust after reviewing outcomes and false positives.

The brief's direct RTO cost is ■65 forward shipping + ■65 return shipping + ■25 packaging/handling + 8% × ■180 expected damage = **■169.40 per RTO**. With 28% COD RTO and 4% prepaid RTO, a successfully converted order has an approximate expected direct-cost reduction of (28% − 4%) × ■169.40 = **■40.66**. A 3% discount on the ■649 average order is ■19.47 (capped at ■30), leaving about ■21.19 expected direct-cost benefit if the customer would otherwise have had average COD risk and completes prepaid payment. This is a scenario, not a guaranteed saving: conversion rate, incremental discount cost, and selection effects matter.

Marketing's acquisition concern matters too. A genuinely cancelled order at ■649 loses an estimated 62% gross contribution, or about ■402, before considering the already-spent acquisition cost. Since the brief says new-customer acquisition can exceed ■300, automatic cancellation can destroy more value than the ■169.40 direct RTO cost. Confirmation and prepaid nudges preserve the chance to complete the sale. Measure this trade-off by new-vs-returning customer segment and acquisition source.

Retrospective action replay. For the latest full calendar month in the CSV (September 2026), 334 labeled rows are present. Applying the initial thresholds produced 30 "ship as-is", 200 "ship after confirmation", 104 "nudge to prepaid", and 0 "hold/cancel" recommendations (auto-cancel is off). This is an exploratory in-sample replay because the demo model is refit on all labeled rows before scoring the historical set; it is not a fair estimate of future performance or savings. Use a later untouched cohort for evaluation.

What could go wrong—and how we would notice

The sample is small and has duplicate IDs; one holdout metric can be unstable; a pincode or courier can change over time; customers may ignore messages; and an address heuristic may unfairly flag a legitimate rural/Hinglish address. Track action counts, delivery/RTO by action and customer segment, confirmation response, prepaid conversion, cancellation reversal requests, complaints, and net contribution. Audit a random sample of flagged orders weekly. Never block an entire pincode without enough data, confidence intervals, and a human appeal path.

What I would build in the next two weeks

1) Clean duplicate/event records and build a later time-based validation set. 2) Connect Shopify order webhooks and create a review queue with auditable reasons. 3) Integrate an approved WhatsApp provider with consent/opt-out handling and idempotent reply webhooks. 4) Verify prepaid payment via a server-side payment webhook. 5) Run a controlled pilot (holdout/control where feasible) and report incremental net contribution—not just RTO reduction. 6) Only then consider an LLM for ambiguous multilingual address interpretation, with strict structured output, abstention and human review.

Scope note: figures above are calculated from the supplied CSV and assignment brief. The app is an offline prototype, not connected to Shopify, a courier, WhatsApp, or payment systems. Model scores and dashboard savings are decision-support estimates, not causal claims.